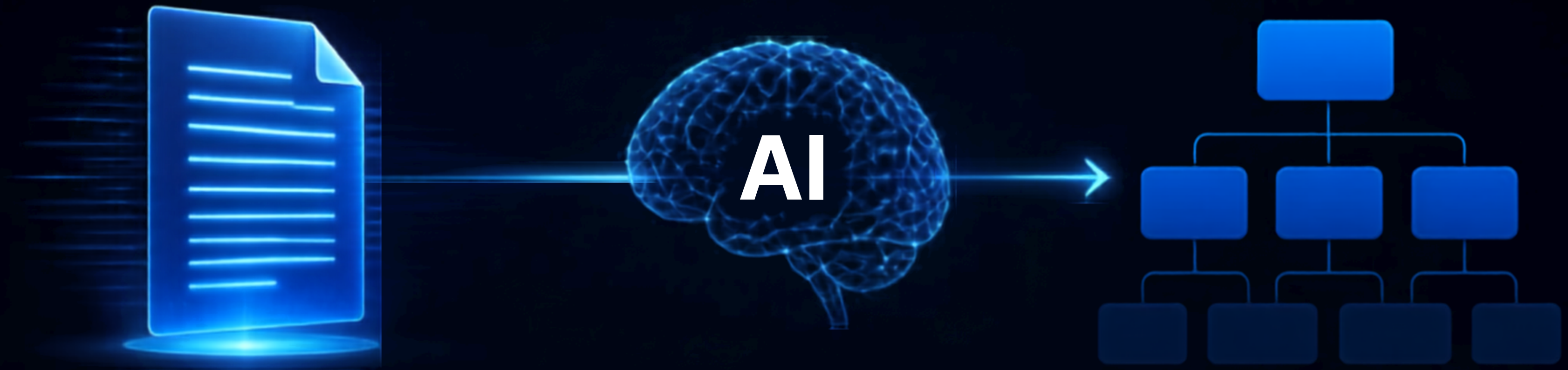




CourseGenie AI

From Syllabus to Complete Course — Instantly

TeachNexa

IBM Bob AI Innovation Hackathon | AI Track

Dhruti Movaliya • Riya Kalariya • Hardi Patel • Ishan Kachhadiya

The Syllabus Is Just the Beginning — Not the End

TIME

1–3 hours per topic
writing materials each semester

FRAGMENTATION

Inconsistent content delivery

REPETITION

Rebuilding materials every semester,
no reuse

$$\begin{array}{ccccccc} 5 & \times & 4 & = & 20 & \times & 2 = 40-100 \\ \text{modules} & & \text{topics} & & \text{topics} & & \text{hrs avg} & & \text{hrs lost} \end{array}$$

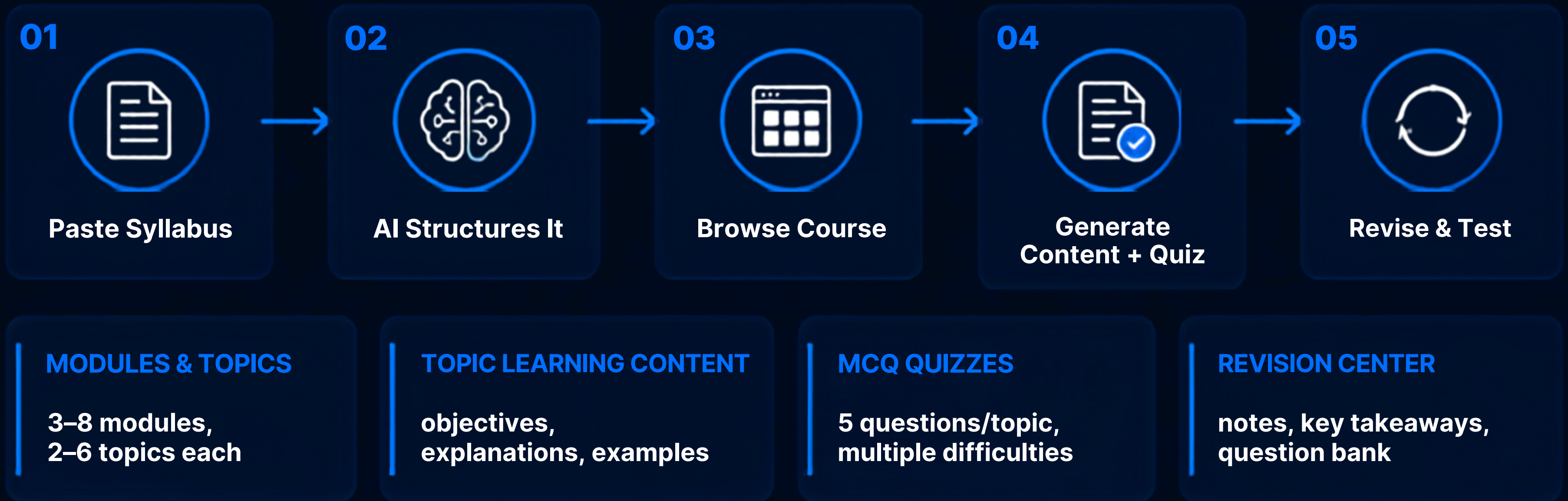
before teaching even begins



AFFECTED

College professors managing 2–4 courses
No instructional design support

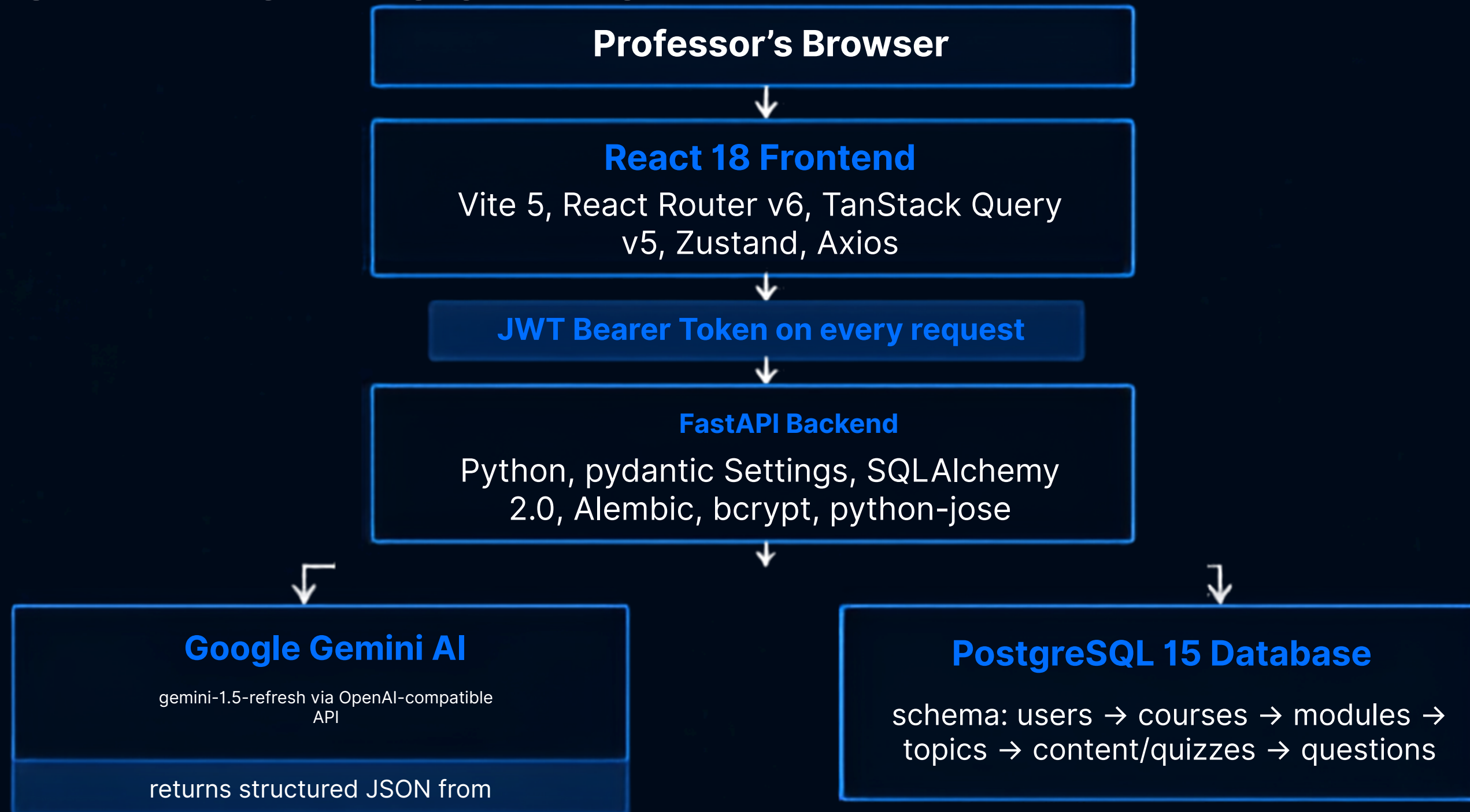
CourseGenie AI — One Input. Complete Course. **Instantly.**



Built with Google Gemini AI + FastAPI + React 18 + PostgreSQL. Fully authenticated. Every professor's course is private and owned by them.

How CourseGenie AI Works

— System Architecture



01

Owner ID assigned
server-side from JWT
— no forgery

02

Regeneration uses
atomic DELETE +INSERT
transaction — no duplicates

03

In-memory SQLite
for CI tests — no
DB dependency

End-to-End in 4 Clicks

1

Sign Up &
Create Course

Syllabus Input

Paste your syllabus

Course Title: Introduction to Data Science

Week 1: Data Basics and Python

Week 2: Statistics Fundamentals

Week 3: Data Visualization

Generate

Paste any syllabus.
Click Generate.
Takes 5–30 seconds.

2

Browse the
AI-Generated Structure

Course Structure

- Module 1: Data Basics and Python
 - Topic 1: What is Data?
 - Topic 1: Python Essentials
- Module 2: Statistics Fundamentals
- Module 3: Data Visualization
- Module 4: Machine Learning Basics

AI decomposes syllabus
into 3–8 modules
with 2–6 topics each.

3

Generate Topic
Learning Content

Topic 1:2: Python Essentials

- Objectives
 - Learning Objectives
- Explanation
- Key Concepts
- Examples

Per-topic: objectives, multi-paragraph
explanation, key concepts, examples.

4

Take the
Quiz & Revise

Quiz

Question 2 of 5

Summary Notes

Option A

Key Formulas

Option B

Cheat Sheet

Option C

Practice Questions

Mind Map

Medium

Next

Revision Center

Summary Notes

Key Formulas

Cheat Sheet

Practice Questions

Mind Map

5-question quiz
with difficulty levels + course-wide
revision materials.

IBM Bob Was Our AI Development Partner — Not Just a Tool

IBM Bob

HOW WE USED IBM BOB

Code generation of FastAPI routes,
Pydantic schemas, models, React components

Architecture review: idempotent content
replacement & JWT auth design

Debugging & refactoring assistance:
SQLAlchemy, JWT timing, React Query cache

Test suite co-ownership:
11 test files, 148+ tests covering auth & ownership

IMPACT OF USING IBM BOB



11 test files written with Bob



148+ tests passing



Full-stack app built in hackathon timeframe



16 REST API endpoints implemented



JWT auth + bcrypt security layer
with **zero** vulnerabilities

“Bob wasn’t a code autocomplete — it was the co-engineer that review architecture, caught edge cases, and helped us ship a production-quality codebase.”

What We Shipped — and What It Means

16

REST API
endpoints live

148+

Tests passing
(11 test files)

4

AI generation methods
(structure, content,
quiz, revision)

3

Database migrations
(full schema evolution)

Structured course hierarchy in **15 seconds** (5 modules × 4 topics)

Full topic content **~10 minutes** for 20 topics)

20 topic quizzes (100 MCQs) in **~10 minutes**

Course-wide revision pack in **30 seconds**

Total time saved: **60–100 hours** per course per semester

LMS Integration

Moodle, Canvas,
Google Classroom



**Student Progress Tracking
& Adaptive Learning**



**Multi-language
Course Generation**

Team TeachNexa

IBM Bob AI Innovation Hackathon • AI Track



Dhruti Movaliya

Team Lead & Database & API
Integration



Riya Kalariya

Frontend Developer



Hardi Patel

Backend Developer



Ishan Kachhadiya

Documentation

CourseGenie AI — Giving professors time back to teach.

<https://github.com/DhrutiM39/bob-ai-hackathon-TeachNexa>